Project 11

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Semantic segmentation of hyperspectral images using foundation models and self-supervised learning

This project deals with semantic segmentation of hyperspectral images [1]. In this context, semantic segmentation corresponds to the process of classifying each pixel into meaningful classes using the rich spectral information contained in the hundreds of contiguous bands of hyperspectral images.
Despite its wide applicative interest, one of the difficulties for automatic semantic segmentation is the limited number of annotated datasets for neural network training. Consequently, this project aims at building on two complementary approaches: 1) leveraging recent foundation models dedicated to hyperspectral images [2] to perform a first 0-shot extraction of features enabling, or at least facilitating, semantic segmentation; 2) using self-supervised strategies such as Barlow twins [3], which enable to learn useful features for the subsequent segmentation and limiting the required number of labeled data. In particular, we will use recent approaches which have been designed for biomedical data.

[1] Grewal, R., Kasana, S. S., & Kasana, G. (2023). Hyperspectral image segmentation: a comprehensive survey. *Multimedia Tools and Applications*, *82*(14), 20819-20872.

[2] Wang, D., Hu, M., Jin, Y., Miao, Y., Yang, J., Xu, Y., ... & Zhang, L. (2025). HyperSIGMA: Hyperspectral intelligence comprehension foundation model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

[3] Zbontar, J., Jing, L., Misra, I., LeCun, Y., & Deny, S. (2021, July). Barlow twins: Self-supervised learning via redundancy reduction. In *International conference on machine learning* (pp. 12310-12320). PMLR.